Course Name: Computer Organization

Course Outcome

CO1- Understand the basics of digital computer system.
CO2- Demonstrate the principle of arithmetic operations on unsigned, signed integers and floating point numbers.
CO3- Understand the concepts of combinational and sequential circuits and their applications.
CO4- Understand the CPU architecture and organization.
CO5- Explain the basic concepts of pipelining.
CO6- Design the steps for the execution of the complete instruction for hardwired and micro-programmed control unit.
CO7- Explain the function of memory hierarchy.
CO8- Determine the interface of CPU with input/output devices and their modes of transfer.

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End Term Examination, Odd Semester 2024-25
B.Tech (CS/CCV/CSF/DA/AIML&IoT/BCT), II year, III Semester
Computer Organization (BCSC 1005)

Time: 3 Hours **Maximum Marks: 50**

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	A general register-organized CPU has 16 registers with 32 bits in each, an ALU, a destination decoder, two data buses (A bus & B bus), and two multiplexers (MUX A & MUX B) (a) How many multiplexers are in the A bus and what is the size of each multiplexer? (b) How many selection inputs are needed for MUX A and MUX B? (c) How many inputs and outputs are there in the decoder? (d) Formulate a control word for the system assuming the ALU has 32 operations.	4	4	U	C
2	Explain PUSH and POP operation. Show how a computer processor evaluates the following arithmetic expression using Reverse Polish Notation? 2+3*(2^5-6)/(7+8*9)-10 **^ is the exponent symbol**	4	4	A	P
3	An instruction is stored at location 300 with its address field at location 301. The address field has the value 400. A processor register R_1 contains the	4	6	U	C

<table>
<tr><td></td><td>number 200. Evaluate the effective address if the addressing mode of the instruction is
(a) Immediate (b) Relative (c) Register indirect (d) Index with R_1 as the index register</td><td></td><td></td><td></td><td></td></tr>
<tr><td>4</td><td>In a certain scientific computation, it is necessary to perform the following arithmetic expression.
$\left(\frac{A_i B_i}{C_i}\right) + D_i * E_i$
(a) Specify the pipeline configuration to carry out the above task.
(b) List the content of all registers for 4 tasks.</td><td>4</td><td>5</td><td>U</td><td>P</td></tr>
<tr><td>5</td><td>A virtual memory has a page size of 1K words. There are eight pages and four blocks. The memory page table contains the following entries:
<table><tr><td>Page</td><td>Block</td></tr><tr><td>0</td><td>3</td></tr><tr><td>1</td><td>1</td></tr><tr><td>4</td><td>2</td></tr><tr><td>6</td><td>0</td></tr></table>
Make a list page-wise of the range of virtual addresses (in decimal) that will cause a page fault if used by the CPU.
OR
Consider a cache memory consisting of 64 blocks of 64 words each which makes a total of 4096 words. Also, assume that the main memory is addressable by a 16-bit address and accommodates 12-bit data. How many bits exist in Tag, Index, Block, and Word fields for the 4-way set associative cache memory? Include a valid bit. What is the size of the cache memory?</td><td>4</td><td>7</td><td>U</td><td>P</td></tr>
</table>

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Why does DMA have priority over the CPU when both request a memory transfer?	3	8	R	F
7	Write the sequence of micro-operations a computer system performs to execute the register-reference instruction CIR (circular shift right). Assume that the 12-bit address field of a 16-bit IR carries	3	6	U	P

	000000000111.				
8	A digital computer has a memory unit of 64K×16 and a cache memory of 1K words. The cache uses direct mapping with a block size of four words. (a) How many bits are there in the tag, index, block, and word fields of the address format? (b) How many bits are there in each word of cache? (c) How many blocks can the cache accommodate?	3	7	A	P
9	Explain different methods of asynchronous data transfer.	3	8	R	F
10	A non-pipeline system takes 50ns to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of 10ns. (a) Determine the speedup ratio of the pipeline for 100 tasks. (b) What is the maximum speedup that can be achieved? (c) Determine the number of clock cycles that it takes to process 200 tasks.	3	5	A	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	A virtual memory system has an address space of 12K words, a memory space of 8K words, and page & block sizes are of 2K words. The following page reference changes occur during a given time interval. 1 0 2 4 2 1 5 3 5 1 2 4 0 2 4 3 Determine the pages that are resident in main memory after each page reference change if the replacement algorithm used is (a) FIFO (b) LRU Also, specify the number of page faults that occurred for each algorithm.	5	7	A	P
12	A computer employs a RAM chip of 512 bytes and a ROM chip of 2048 bytes. Determine the number of RAM and ROM chips required to fulfill the capacity of 4K bytes of RAM and 8K bytes of ROM. If 16 address lines of the CPU are mapped to RAMs and ROMs then, compute and show the	5	7	U	C

	address range of each component in hexadecimal.				
13	The initial content of PC is 7FF. The content of memory at address 7FF is 9A9F. The content of memory at address A9F is 0C35. The content of memory at address C35 is FFFF. List the contents in hexadecimal of registers Program Counter (PC), Address Register (AR), Data Register (DR), and Accumulator (AC) of the basic computer when an ADD instruction is fetched from memory and executed.	5	6	U	C